

Chapter 1 Einf

- (1-Dimensional)
- (Wave)
- (Heat Core)
- (Schrödinger)
- (Burger's Equation)
- (Minimal Area)
- (Navier-Stokes' Equation)
 - ($\mu \neq 0$)
 - ($\mu = 0$, Euler Equation)
- (KortewegDe Vries (KdV) equation)

1.1 Wärmeleitung Problem

Given:

- Region $\Omega \subseteq \mathbb{R}^3$;
- Masse $u : \Omega \times I \rightarrow \mathbb{R}$
- Flussschade $j : \Omega \times [0, T] \rightarrow \mathbb{R}^3$;

Prinzip

- $\forall x \in \Omega$,

$$\begin{aligned} \frac{d}{dt} \int_{B_r(x)} u(-, t) dy &= - \int_{\partial B_r(x)} j \cdot \nu dS + \int_{B_r(x)} f(-, t) dy \\ &\Rightarrow \int_{B_r(x)} (\partial_t u + \operatorname{div}(j) - f) dy = 0 \\ &\Rightarrow \partial_t u + \operatorname{div}(j) + f = 0 \end{aligned}$$

Especially, $u \text{ const} \Rightarrow$

1.2 Harmonic Equations

Definition 1.2.0.1 $\Omega \subseteq \mathbb{R}^n$ open. $u \in C^2(\Omega)$. u is harmonic $\Leftrightarrow -\Delta u = 0$

♣ Example 1.2.0.2

- $u(x) = Ax + b$;
- Quadrics $A \in \mathbb{R}^{n \times n}$, $u(x) := xAx^T = \sum_{i,j=1}^n x_i a_{ij} x_j \Rightarrow D^2 u = A, \Delta u = \operatorname{tr}(d^2 u) = \operatorname{tr}(A)$, so u is harmonic $\Leftrightarrow \operatorname{tr}(A) = 0$;

- Variables $n = 2$, $u(x_1, x_2) = f(x_1)g(x_2)$,

$$\Delta u =$$

- Holomorphic Function $\Omega \subseteq \mathbb{C}$, harmonic $\Leftrightarrow u, v$ harmonic.

$$\Delta f = \Delta u + i\Delta v$$

1.3 Approximation through Convolution

Definition 1.3.0.1 $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$(f * g)(x) := \int_{\mathbb{R}^n} f(y-x)g(x)$$

◇ **Lemma 1.3.0.2** (Convolution Approximation) $f \in L^p(\mathbb{R})$, $1 \leq p \leq \infty$, $g \in$

◦ **Proof**

- $p = 1$:
- $p = \infty$:
- $p \in (1, \infty)$:

□

◇ **Lemma 1.3.0.3** $f \in L^1(\mathbb{R}^n)$, $f = 0$ in $\mathbb{R}^n$ or B for

$$g \in L^1_\infty(\mathbb{R}^n) \Rightarrow f * g \in$$

If

◦ **Proof ...**

□

◇ **Lemma 1.3.0.4** Suppose (η) standard Dirac-sequence, then

...

1.4 Regularity

◇ **Lemma 1.4.0.1** Given a mollification η , $\eta_r(x) := r^{-n}\eta(\frac{x}{r})$ $u \in C^2(\Omega)$ and $u(x) = \oint_\partial B dS \Rightarrow u(x) = (u * \eta_r)(x), \forall \bar{B}_r(x) \subseteq \Omega$

◦ **Proof ...**

□

♠ **Proposition 1.4.0.2** $\Omega \subseteq \mathbb{R}^n$ open, $U \in C^0(\Omega)$. If $u(x) = \int_{B_r(x)} u dy$, then $u \in C^\infty(\Omega)$, $\Delta u = 0$.

◦ **Proof ...**

□